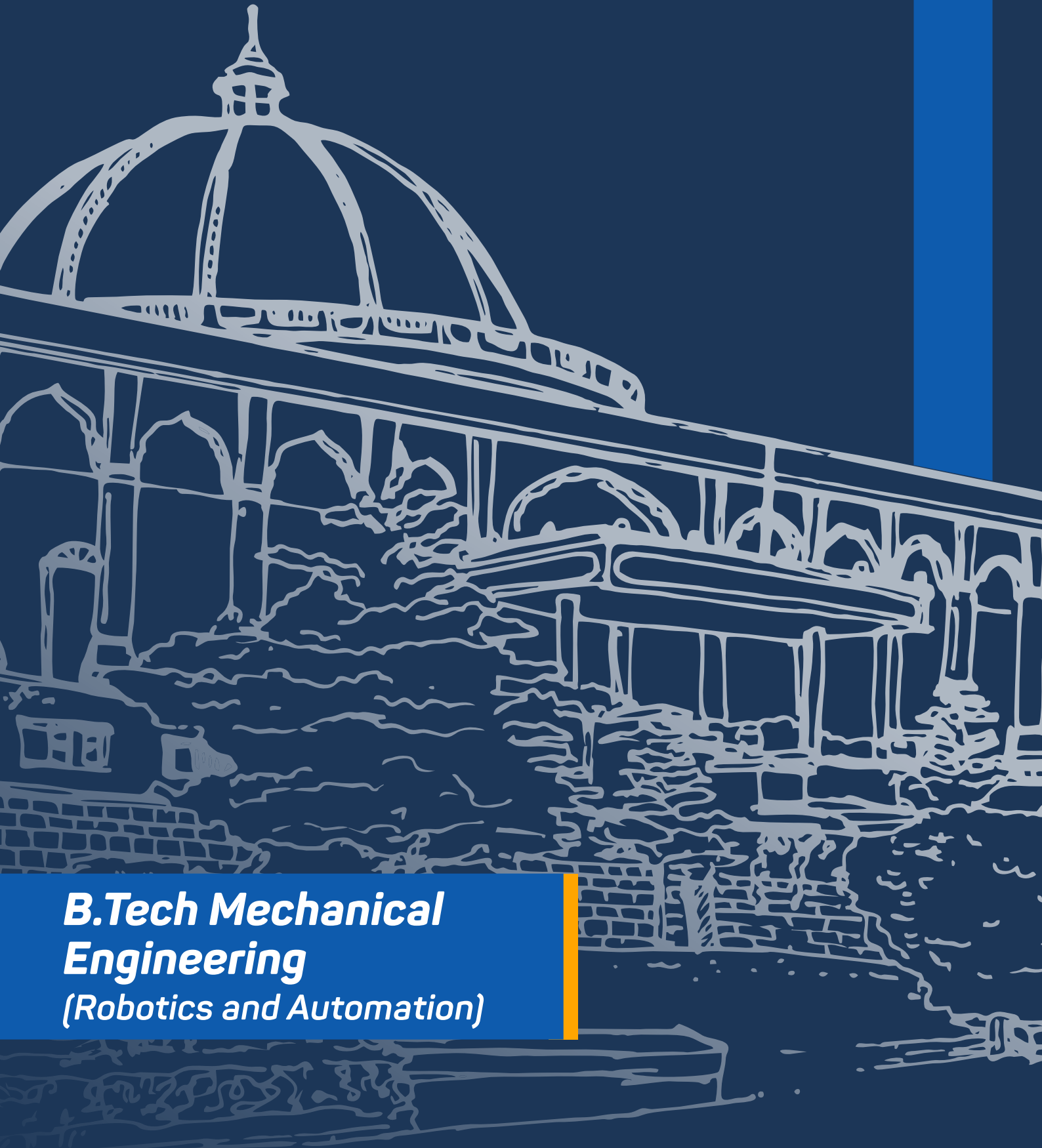




***B.Tech Mechanical
Engineering
(Robotics and Automation)***

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	Department of Mechanical Engineering
Programme Name	B.Tech. Mechanical Engineering (Robotics and Automation)

Semester	Odd (I)	Even (II)	Total Credits
First Year	22	21	43
Second Year	21	22	43
Third Year	22	22	44
Fourth Year	21	14	35

+ + + + + + + + + + + + + + + **COURSE BASKET** + + + + + + + + + + + + + + +

| Course Type | Description |
|----------------------|--|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU |
| Programme Electives | Open electives under the programme allow students to specialise in a particular area connected to their major. |
| University Core | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core. |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| I | UC | Indian Constitution | 1 |
| I | UC | Environment and Sustainability | 1 |
| I | UC | Yoga - I | 1 |
| I | UC | Social Leadership Development Program | 1 |
| I | UC | Financial Literacy | 1 |
| I | PF | Linear Algebra and Differential Calculus | 3 |
| I | PF | Engineering Physics | 3 |
| I | PF | Engineering Chemistry | 3 |
| I | Program Capstone Project/Seminar and Internships | Ideas and Innovations in Manufacturing | 1 |
| I | PF | Biology for Engineers | 2 |
| I | PF | Engineering Graphics | 3 |
| I | Program Capstone Project/Seminar and Internships | Computer Aided Modelling | 2 |
| | | Total Credits: | 22 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| II | UC | Yoga - II | 1 |
| II | UC | Co-creation | 1 |
| II | UC | AI for everyone | 2 |
| II | UC | Foundation of Peace | 2 |
| II | UC | Indian Knowledge System (General) | 2 |
| II | UC | Sports | 1 |
| II | PF | Integral Calculus | 3 |
| II | PF | Introduction to Electrical and Electronics Engineering | 3 |
| II | PF | Programming and Problem Solving | 3 |
| II | PF | Engineering Mechanics | 3 |
| | | Total Credits: | 21 |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| III | UC | Spiritual and Cultural heritage: Indian Experience | 2 |
| III | UC | Research Innovation Design Entrepreneurship (RIDE) | 1 |
| III | UE | University Electives - I | 3 |
| III | PF | Engineering Mathematics | 4 |
| III | PM | Introduction to Robotics | 4 |
| III | PM | Mechanics of Materials | 3 |
| III | PM | Basic IoT Lab | 1 |
| III | PM | Material Science & Engineering Metallurgy | 3 |
| | | Total Credits: | 21 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| IV | UC | Rural Immersion | 1 |
| IV | UC | Life Transformation Skills | 1 |
| IV | UE | University Electives - II | 3 |
| IV | PM | Manufacturing Science | 4 |
| IV | PF | Indian Knowledge System (Sci. & Tech.) | 2 |
| IV | PF | Robot Kinematics & Dynamics | 4 |
| IV | PM | Design of Machine Elements | 3 |
| IV | Program Capstone Project/
Seminar and Internships | Robotics Engineering Software Lab | 1 |
| IV | PM | Thermo-Fluids | 3 |
| | | Total Credits: | 22 |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| V | UC | Managing Conflicts Peacefully: Tools and Techniques | 2 |
| V | UE | University Electives - III | 3 |
| V | PE | Program Elective - I | 4 |
| V | PF | Data Science for Engineers | 2 |
| V | PM | Robot Drive Systems and Programming | 4 |
| V | PM | Hydraulics and Pneumatic Systems | 4 |
| V | PF | Microcontroller and Applications | 3 |
| | | Total Credits: | 22 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| VI | UC | National Academic Immersion Program | 2 |
| VI | PE | Program Elective - II | 4 |
| VI | PM | Industrial Automation | 4 |
| VI | PM | Artificial Intelligence and Machine Learning | 3 |
| VI | PM | Robot Control System | 3 |
| VI | Program Capstone Project/Seminar and Internships | Seminar | 1 |
| VI | Program Capstone Project/Seminar and Internships | Mini Project Lab | 1 |
| VI | PM | Robot Operating System | 4 |
| | | Total Credits: | 22 |

Semester - VII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|-----------------------------|---------------|
| VII | PE | Program Elective - III | 4 |
| VII | PE | Program Elective - IV | 4 |
| VII | PM | Balancing and Transmission | 3 |
| VII | Program Capstone Project/Seminar and Internships | Capstone Project | 8 |
| VII | PM | Robot Vision and Perception | 2 |
| | | Total Credits: | 21 |

Semester - VIII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|----------------------------|---------------|
| VIII | Program Capstone Project/Seminar and Internships | Internship | 14 |
| | | Total Credits: | 14 |

Program Electives

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|------------------------|---|---------------|
| V | Program Elective - I | Manipulator Design | 4 |
| V | Program Elective - I | Embedded Systems and RTOS | 4 |
| V | Program Elective - I | Additive Manufacturing | 4 |
| V | Program Elective - I | Mechatronics Systems | 4 |
| V | Program Elective - I | PLC and SCADA | 4 |
| VI | Program Elective - II | Mechanical Vibrations | 4 |
| VI | Program Elective - II | Sensor Technology | 4 |
| VI | Program Elective - II | Metrology and Quality Control | 4 |
| VI | Program Elective - II | Mobile Robots and AGV | 4 |
| VI | Program Elective - II | Finite Element Method | 4 |
| VII | Program Elective - III | Autotronics | 4 |
| VII | Program Elective - III | Industrial Robotics | 4 |
| VII | Program Elective - III | Industry 5.0 | 4 |
| VII | Program Elective - III | Unmanned Aerial Vehicles | 4 |
| VII | Program Elective - III | Underwater Robotics | 4 |
| VIII | Program Elective - III | Maintenance & Safety in Robotics | 4 |
| VIII | Program Elective - IV | Intelligent Robotic Systems | 4 |
| VIII | Program Elective - IV | Computer Integrated Manufacturing Systems | 4 |
| VIII | Program Elective - IV | Project Management and Operation Research | 4 |
| VIII | Program Elective - IV | Augmented Reality and Virtual Reality | 4 |
| VIII | Program Elective - IV | Automotive and E-Vehicle Technology | 4 |
| VIII | Program Elective - IV | Micro electro mechanical System | 4 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.